CSM
Spring 2026

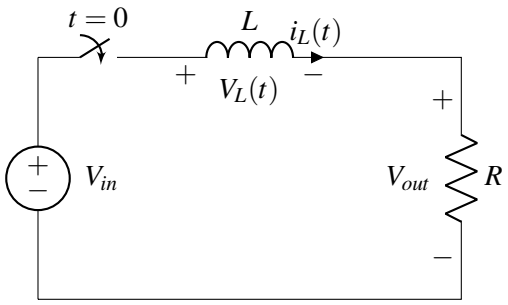
EECS 16B
Worksheet 6

1. Introduction to Inductors

Now that we are comfortable solving for the transient behavior of charging and discharging capacitors, we can move to analyzing a new circuit element: the inductor. An inductor has the physical property of *inductance*, represented by a constant L. Inductors are characterized by the following I-V relationship:

$$V_L(t) = L\frac{d}{dt}i_L(t) \tag{1}$$

Let's analyze the following LR circuit, with the initial condition $i_L(0) = 0$ with the switch open before $t = 0$:



(a) **Write out the differential equation for the current $i_L(t)$ of the inductor starting at $t = 0$ when the switch is closed.**

Solution: KCL and Ohm's law gives us:

$$\begin{aligned} V_{in} - V_{out} = V_L(t) &= L\frac{d}{dt}i_L(t) \\ V_{out} &= Ri_L(t) \\ \implies L\frac{d}{dt}i_L(t) &= V_{in} - Ri_L(t) \\ \implies \frac{d}{dt}i_L(t) &= -\frac{R}{L}i_L(t) + \frac{V_{in}}{L} \end{aligned}$$

(b) **Solve the differential equation for $i_L(t)$.**

HINT: given a first order differential equation $\frac{dy(t)}{dt} + \alpha y(t) = \beta$, the solution is:

$$y(t) = y(0)e^{-\alpha t} + \frac{\beta}{\alpha}(1 - e^{-\alpha t}) \tag{2}$$

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Solution:

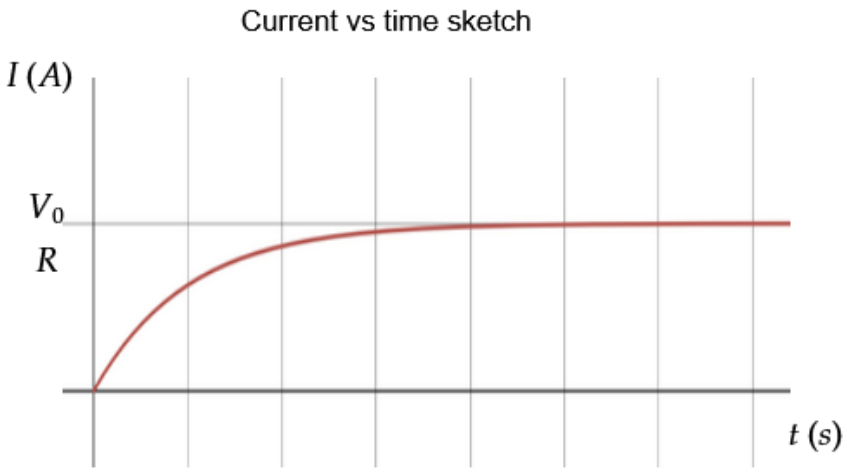
$$\frac{d}{dt}i_L(t) = -\frac{R}{L}i_L(t) + \frac{V_{in}}{L}$$
$$\implies i_L(t) = \frac{V_{in}}{R}(1 - e^{-\frac{R}{L}t})$$

(c) **What is the steady-state current through the inductor as $t \to \infty$?** Sketch a plot of the current through the inductor over time, labeling the asymptote after reaching the steady-state. This should provide you with some intuition as to the physical behavior of an inductor once an inductor is at steady state.

Solution:

$$\lim_{t\to\infty} i_L(t) = \lim_{t\to\infty} \frac{V_{in}}{R}(1 - e^{-\frac{R}{L}t})$$
$$= \frac{V_{in}}{R}(1 - 0)$$
$$= \frac{V_{in}}{R}$$

The physics of the inductor opposes current flow initially, but it approaches the steady-state current as determined by the rest of the circuit (the resistor and voltage source).



(d) **What is the steady-state voltage drop across the inductor as $t \to \infty$?**

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Solution:

$$\begin{aligned}
\lim_{t\to\infty} V_L(t) &= \lim_{t\to\infty} L\frac{d}{dt} i_L(t) \\
&= \lim_{t\to\infty} L\left(\frac{d}{dt}\frac{V_{in}}{R}(1-e^{-\frac{R}{L}t})\right) \\
&= \lim_{t\to\infty} L\frac{V_{in}}{R}\frac{R}{L}e^{-\frac{R}{L}t} \\
&= \lim_{t\to\infty} V_{in}e^{-\frac{R}{L}t} \\
&= 0
\end{aligned}$$

(e) **What circuit element does the inductor act like at steady-state?**

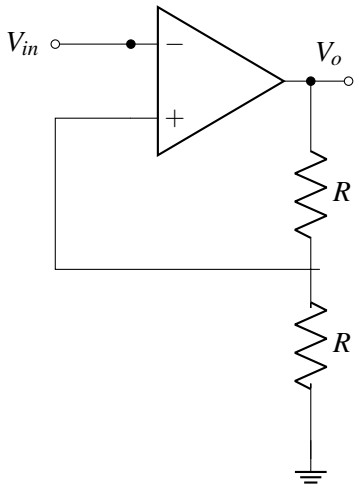
Solution: It acts like a wire element, since the voltage drop across the inductor goes to 0.

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2. OpAmps as Amplifiers

(a) In general, what is feedback? What are positive and negative feedback? And how do you introduce feedback into op-amps?

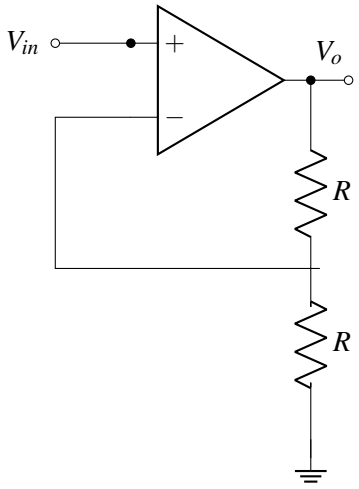
Solution: Feedback is the process by which an output feeds back into an input. Positive feedback is where a positive or negative output pours more positive or negative (respectively) into the input and causes a loop of growth (or shrinkage); think of momentum, where you put in work to move something, and then that movement causes that thing to keep moving faster. Negative feedback is where a positive or negative output introduces a negative or positive (respectively) input that tries to balance the output. Think of pushing a spring, you exerting energy will cause the spring to try to fight back in the other direction and balance out the energy you are putting in.
In an op-amp, we get a feedback loop if the output connects back to the input.



(b) Referring to the above example of an op-amp employing positive feedback, start with the assumption that $V_+ = V_-$. Now, taking into account the fact that $V_{out} = A(V_+ - V_-)$ (where A is some gain), try to raise V_{out} (by messing with V_{in}) and observe what happens. Try to lower V_{out} and observe that too.

Solution: When we raise V_{out}, for example by lowering $V_{in} = V_-$, we see that the now positive output comes back and is also raising V_+, leading to an even more positive output, which in turn leads to an even more positive output...until the op-amp eventually hits its rail. When we lower V_{out} by raising V_{in}, our now negative V_{out} feeds back into V_+, leading to an even more negative output, which leads to an even more negative output, etc.

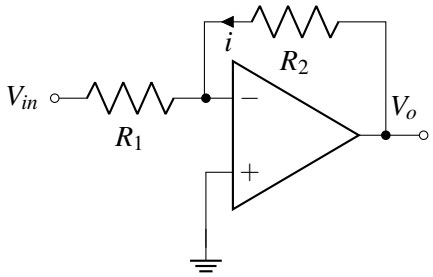
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(c) Now do the same with this above example of an op-amp that uses negative feedback

Solution: This time when we raise V_{out}, in this case by raising $V_{in} = V_+$, we get an increase to V_-, causing the difference $(V_+ - V_-)$ to shrink. If the difference is still positive then it will shrink again, if our V_- ever overshoots and causes the whole thing to become negative, then $(V_+ - V_-)$ becomes positive and raises V_- until we are at an equilibrium, that is where $V_+ = V_-$. Lowering V_{out} by lowering V_{in} has the same equalizing effect.

The takeaway from this is that in the case of negative feedback, there is no way for V_+ to not equal V_-, since if it does for a moment it will end up balancing back out. This is where we get that golden rule of negative feedback that $V_+ = V_-$!



(d) In terms of the input V_{in}, what is V_{out} for this circuit? What is the gain of this circuit?

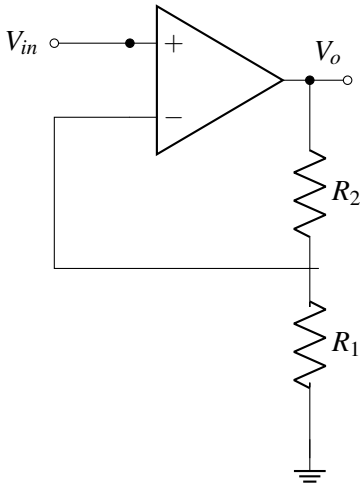
Solution: We can use the golden rules of op-amps and nodal analysis to find V_{out} as a function of V_{in}. Since V_+ is connected to ground, and the golden rules state that $V_+ = V_-$, we can conclude that $V_- = 0V$. In addition, since $I_+ = I_- = 0$, we know there is no current going into the op-amp, so all of the current from V_{in} goes to R_2. Doing nodal analysis on V_+,

$$\frac{0 - V_{in}}{R_1} = \frac{V_{out} - 0}{R_2}$$

$$V_{out} = -V_{in}\frac{R_2}{R_1}$$

The gain is how much V_{out} scales V_{in}, which is $-\frac{R_2}{R_1}$. This particular type of op-amp is called an *inverting* amplifier since you can use this op-amp to negate and scale V_{in} to the output you want.

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(e) Again, in terms of the input V_{in}, what is V_{out} for this circuit? What is the gain?

Solution: Like we did in part (d), we can use the golden rules of op-amps and nodal analysis to find V_{out} as a function of V_{in}. We know that $V_+ = V_-$ because of the golden rules, and that $V_+ = V_{in}$. In addition, since $I_+ = I_- = 0$, all of the current from V_{out} goes to ground. Doing nodal analysis on V_-,

$$\frac{V_{in} - 0}{R_1} = \frac{V_{out} - V_{in}}{R_2}$$

$$V_{out} = V_{in}\frac{R_2}{R_1} + V_{in} = V_{in}(\frac{R_2}{R_1} + 1)$$

The gain in this case is $\frac{R_2}{R_1} + 1$. This particular type of op-amp is called a *non-inverting* amplifier, since with careful choice of R_1 and R_2, you can use this op-amp to scale V_{in} to the output you want.

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3. Fun with Inductors (Challenge)

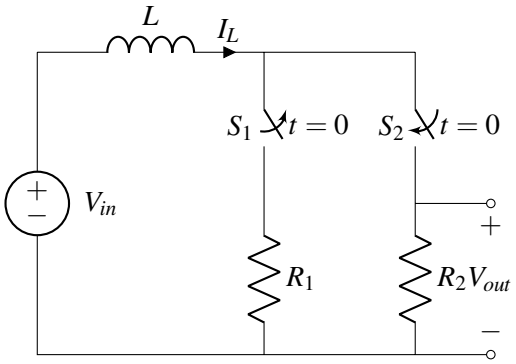


Figure 1: Circuit A

(a) Consider circuit A. Assuming that for $t < 0$, switch S_1 is on and switch S_2 is off (and both switches have been in these states indefinitely), what is $i_L(0)$?

Solution: When S_1 is on and S_2 is off for a long period of time, $\frac{di_L}{dt} = 0$ because the circuit will have reached a steady state, and the current through R_1 will be equal to i_L. We find

$$V_{in} - V_L - V_{R_1} = 0$$

$$V_{in} - L\frac{di_L}{dt}(0) - i_L(0)R_1 = 0$$

$$V_{in} - i_L(0)R_1 = 0$$

$$i_L(0) = \frac{V_{in}}{R_1}$$

(b) Now let's assume that for $t \geq 0$, S_1 is off and S_2 is on. Solve for $V_{out}(t)$ for $t \geq 0$.

Solution:

$$V_{in} - V_L - V_{out} = 0$$

$$V_{in} - L\frac{di_L}{dt} - i_L R_2 = 0$$

$$\frac{di_L}{dt} + \frac{R_2}{L}i_L = \frac{V_{in}}{L}$$

This is a non-homogenous first order differential equation in i_L. We can solve for $i_L(t)$ and then use Ohm's law to find $V_{out}(t)$ after this has been solved.

$$\frac{di_L}{dt} + \frac{R_2}{L}(i_L - \frac{V_{in}}{R_2}) = 0$$

Let $\tilde{i_L} = i_L - \frac{V_{in}}{R_2}$. We now have:

$$\frac{d\tilde{i_L}}{dt} + \frac{R_2}{L}\tilde{i_L} = 0$$

The general solution is given by:

$$\tilde{i_L}(t) = c_1 e^{-\frac{R_2}{L}t}$$

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Resubstituting back i_L, we have:

$$i_L(t) = \frac{V_{in}}{R_2} + c_1 e^{-\frac{R_2}{L}t}$$

Applying initial conditions, we know:

$$i_L(0) = \frac{V_{in}}{R_2} + c_1 = \frac{V_{in}}{R_1}$$

$$c_1 = \frac{V_{in}}{R_1} - \frac{V_{in}}{R_2}$$

Our solution for $i_L(t)$ thus becomes:

$$i_L(t) = \frac{V_{in}}{R_2} + (\frac{V_{in}}{R_1} - \frac{V_{in}}{R_2})e^{-\frac{R_2}{L}t}$$

Since $V_{out}(t) = i_L(t)R_2$,

$$V_{out}(t) = V_{in}(1 + (\frac{R_2}{R_1} - 1)e^{-\frac{R_2}{L}t})$$

(c) If $V_{in} = 1V$, $L = 1nH$, $R_1 = 1k\Omega$, and $R_2 = 10k\Omega$, what is the maximum value of $V_{out}(t)$ for $t \geq 0$?

Solution: Since the coefficient in front of our time-varying component $e^{-\frac{R_2}{L}t}$, given by $\frac{R_2}{R_1} - 1 = 9$, is positive, $V_{out}(t)$ undergoes decay over time. Therefore, the maximum value is achieved at $t = 0$:

$$\max V_{out}(t) = V_{out}(0) = \frac{R_2}{R_1} V_{in} = 10V$$

(d) In general, if we want $\max V_{out}(t)$ to be greater than V_{in}, what relationship needs to be maintained between the values of R_1 and R_2?

Solution: As long as the coefficient on our exponential term, given by $\frac{R_2}{R_1} - 1$, is greater than 0 (i.e. when $\frac{R_2}{R_1} > 1$) then the maximum value of $V_{out}(t)$ will be achieved at $t = 0$ and will have a value of $\frac{R_2}{R_1} V_{in} > V_{in}$. Otherwise, if $\frac{R_2}{R_1} \leq 1$, the maximum value of $V_{out}(t)$ is reached at $t = \infty$, where $V_{out} = V_{in}$ regardless of R_2 and R_1. Therefore, our necessary condition for the maximum of V_{out} to be greater than V_{in} is:

$$R_2 > R_1$$

(e) Now assume that at time $t = t_1$, switch S_2 was turned off, and switch S_1 was turned back on. Solve for $i_L(t)$ for $t > t_1$. If $R_2 > R_1$, how does this $i_L(t)$ for $t > t_1$ compare with the initial condition $i_L(0)$ you found in part (a)?

Solution: Our new initial condition for $t > t_1$ is given by plugging in $t = t_1$ into the equation for $i_L(t)$ we found in part (b). Thus, $i_L(t_1) = \frac{V_{in}}{R_2} + (\frac{V_{in}}{R_1} - \frac{V_{in}}{R_2})e^{-\frac{R_2}{L}t_1}$.
We can write the relationship between the current through the inductor and the current through R_1:

$$i_L = i_{R_1}$$

$$i_L = \frac{V_{R_1}}{R_1}$$

$$i_L = \frac{V_{in} - V_L}{R_1}$$

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$$i_L = \frac{V_{in}}{R_1} - \frac{L\frac{di_L}{dt}}{R_1}$$

$$\frac{di_L}{dt} + \frac{R_1}{L} i_L = \frac{V_{in}}{L}$$

This is a first order non-homogeneous differential equation similar to that found in part (b), except with R_1 in place of R_2. Following those steps in part (b), we find the general solution:

$$i_L(t) = \frac{V_{in}}{R_1} + c_1 e^{-\frac{R_1}{L}t}$$

To find c_1 we apply our initial condition:

$$i_L(t_1) = \frac{V_{in}}{R_1} + c_1 e^{-\frac{R_1}{L}t_1} = \frac{V_{in}}{R_2} + (\frac{V_{in}}{R_1} - \frac{V_{in}}{R_2})e^{-\frac{R_2}{L}t_1}$$

$$c_1 e^{-\frac{R_1}{L}t_1} = (\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(1 - e^{-\frac{R_2}{L}t_1})$$

$$c_1 = \frac{(\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(1 - e^{-\frac{R_2}{L}t_1})}{e^{-\frac{R_1}{L}t_1}}$$

$$c_1 = (\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(e^{\frac{R_1}{L}t_1} - e^{\frac{R_1 - R_2}{L}t_1})$$

Thus, we have

$$i_L(t) = \frac{V_{in}}{R_1} + (\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(e^{\frac{R_1}{L}t_1} - e^{\frac{R_1 - R_2}{L}t_1})e^{-\frac{R_1}{L}t}$$

for $t > t_1$. We also see that as $t \to \infty$, $i_L(t)$ for $t > t_1$ becomes:

$$i_L(t = \infty) = \frac{V_{in}}{R_1} + (\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(e^{\frac{R_1}{L}t_1} - e^{\frac{R_1 - R_2}{L}t_1})e^{-\frac{R_1}{L}\infty}$$

$$i_L(t = \infty) = \frac{V_{in}}{R_1} + (\frac{V_{in}}{R_2} - \frac{V_{in}}{R_1})(e^{\frac{R_1}{L}t_1} - e^{\frac{R_1 - R_2}{L}t_1})(0)$$

$$i_L(t = \infty) = \frac{V_{in}}{R_1} = i_L(0)$$

Thus, if we turn S_2 back off and S_1 back on as was described in this part, we will eventually revert back to the initial state from which we started! Specifically, if $R_2 > R_1$, $i_L(t)$ at $t = t_1$ will be less than our initial condition $i_L(0)$, and $i_L(t)$ will rise to $i_L(0)$ over time.

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